

To do Hands on:

Use Azure portal environment called the sandbox

Exam weightage:

AZ-900 Domain Area	Weight
Describe cloud concepts	25-30%
Describe Azure architecture and services	35-40%
Describe Azure management and governance	30-35%

Learning Path:

- Define cloud computing.
- Describe the shared responsibility model.
- Define cloud models, including public, private, and hybrid.
- Identify appropriate use cases for each cloud model.
- Describe the consumption-based model.
- Compare cloud pricing models.

Cloud Computing:

Cloud computing is the delivery of computing services over the internet. Computing services include common IT infrastructure such as virtual machines, storage, databases, and networking

Basic service provided by each cloud provider: **Compute(how much processing your computer do) and Storage(volume of data you store in computer)**

Shared Responsibility Model:

- responsibilities get shared between the cloud provider and the consumer.
- Physical security, power, cooling, and network connectivity are the responsibility of the cloud provider
- consumer is responsible for the data and information stored in the cloud. and also access security(you will give access to those who need it)

cloud service types :

1.infrastructure as a service (IaaS)**Azure Virtual Machines, Azure Disk Storage, and virtual networks.,**

2.platform as a service (PaaS) **Azure App Service, Azure Functions, Azure SQL Database, and Azure Storage.,**

3.software as a service (SaaS) **Microsoft 365, Dynamics 365**

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Shared	Shared	Customer	Customer
	Network controls	Shared	Shared	Customer	Customer
	Operating system	Shared	Shared	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

Microsoft	Customer	Shared
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Cloud Models:

- deployment type of cloud resources.

Private Cloud:

- Used by single entity
- Great control over it
- Comes with great cost and fewer of benefits of public cloud deployment
- may be hosted from your on site datacenter.
- hosted in a dedicated datacenter offsite, potentially even by a third party that has dedicated that datacenter to your company.

Public Cloud:

- built, controlled, and maintained by a third-party cloud provider.
- anyone that wants to purchase cloud services can access and use resources.

Hybrid Cloud:

- uses both public and private clouds in an inter-connected environment
- provide an extra layer of security(user can decide which service to use public or private)

Multi-Cloud:

- use multiple public cloud providers.

Azure Arc:

- set of technologies that helps manage your cloud environment.(even it is public,private,hybrid or multi cloud environment)

Azure VMware Solution:

- established with VMware in a private cloud environment but want to migrate to a public or hybrid cloud? Azure VMware Solution lets you run your VMware workloads in Azure with seamless integration and scalability

Consumption-based Model:

- No upfront costs.
- No need to purchase and manage costly infrastructure that users might not use to its fullest potential.
- The ability to pay for more resources when they're needed.
- The ability to stop paying for resources that are no longer needed.

CapEx : one-time, up-front expenditure to purchase or secure tangible resources

OpEx : spending money on services or products over time.

Eg. Cloud computing(using a pay-as-you-go pricing model.)

Additional resources:

[Introduction to Azure hybrid cloud services](#)

[Introduction to Azure VMware Solution](#)

[Shared responsibility model](#)

Learning objectives

After completing this module, you'll be able to:

- Describe the benefits of high availability and scalability in the cloud.
- Describe the benefits of reliability and predictability in the cloud.
- Describe the benefits of security and governance in the cloud.
- Describe the benefits of manageability in the cloud.

Benefits of Cloud computing:

High availability:

- focuses on ensuring maximum availability, regardless of disruptions or events that may occur.
- Azure is a highly available cloud environment with uptime guarantees which is part of SLAs(service level agreements)
- **Note:** Before building architecture diagram check the SLAs(which is in %) for each azure resources because it is crucial for uptime of your application

Scalability:

- refers to the ability to adjust resources to meet demand
- **Vertical scaling** is focused on increasing or decreasing the capabilities of resources.(adding or removing CPU or RAM specifications.)
- **Horizontal scaling** is adding or subtracting the number of resources(add or remove additional virtual machines or containers either automatically or manually)

Reliability :

- ability of a system to recover from failures and continue to function
- With this global scale, even if one region has a catastrophic event other regions are still up and running
- You can design your applications to automatically take advantage of this increased reliability

Predictability:

- Deploy a solution built around this framework and you have a solution whose cost and performance are predictable.

- **Performance predictability:** predicting the resources needed to deliver a positive experience for your customers. **Autoscaling, load balancing, and high availability** are just some of the cloud concepts that support performance predictability.

- **Cost predictability:** focused on predicting or forecasting the cost of the cloud spend.
- use tools like the **Total Cost of Ownership (TCO)** or **Pricing Calculator**

Security and Governance:

- all your deployed resources meet corporate standards and government regulatory requirements.
- **Cloud-based auditing** helps flag any resource that's out of compliance with your corporate standards and provides mitigation strategies.
- cloud providers are typically well suited to handle things like **distributed denial of service (DDoS) attacks**, making your network more robust and secure.
- establishing a good governance footprint early, you can keep your cloud footprint updated, secure, and well managed.

Manageability:

1.Management of the Cloud:

- Managing cloud resource.1.Automatically scale resource deployment based on need.2. Deploy resources based on a preconfigured template, removing the need for manual configuration.3.Monitor the health of resources and automatically replace failing resources.4.Receive automatic alerts based on configured metrics, so you're aware of performance in real time.

2.Management in the Cloud

- how you're able to manage your cloud environment and resources.
 - 1.Through a web portal.
 - 2.Using a command line interface.
 - 3.Using APIs.
 - 4.Using PowerShell.

Additional Resources:[Build great solutions with the Microsoft Azure Well-Architected Framework](#)

Learning objectives

After completing this module, you'll be able to:

- Describe infrastructure as a service (IaaS).
- Describe platform as a service (PaaS).
- Describe software as a service (SaaS).
- Identify appropriate use cases for each cloud service (IaaS, PaaS, SaaS).

Infrastructure as a service (IaaS)

- most flexible category of cloud services, as it provides you the maximum amount of control for your cloud resources.
- cloud provider is responsible for maintaining the hardware, network connectivity (to the internet), and physical security.
- You're responsible for everything else: operating system installation, configuration, and maintenance; network configuration; database and storage configuration
- scenarios where IaaS might make sense include:
 1. **Lift and Shift migration** : setting up cloud resources similar to your on-prem datacenter and migrate to IaaS
 2. **Testing and development**: You have established configurations for development and test environments that you need to rapidly replicate

Platform as a Service(PaaS):

- middle ground between renting space in a datacenter (infrastructure as a service) and paying for a complete and deployed solution (software as a service)
- In a PaaS scenario, you don't have to worry about the licensing or patching for operating systems and databases.
- scenarios where PaaS might make sense include:

1. **Development framework:** PaaS provides a framework that developers can build upon to develop or customize cloud-based applications.

2. **Analytics or business intelligence:** Tools provided as a service with PaaS allow organizations to analyze and mine their data, finding insights and patterns and predicting outcomes to improve forecasting, product design decisions, investment returns, and other business decisions.

Software as a service (SaaS) :

- most complete cloud service model from a product perspective. With SaaS, you're essentially renting or using a fully developed application.
- scenarios where PaaS might make sense include:
 - Email and messaging.
 - Business productivity applications.
 - Finance and expense tracking.

Learning objectives

After completing this module, you'll be able to:

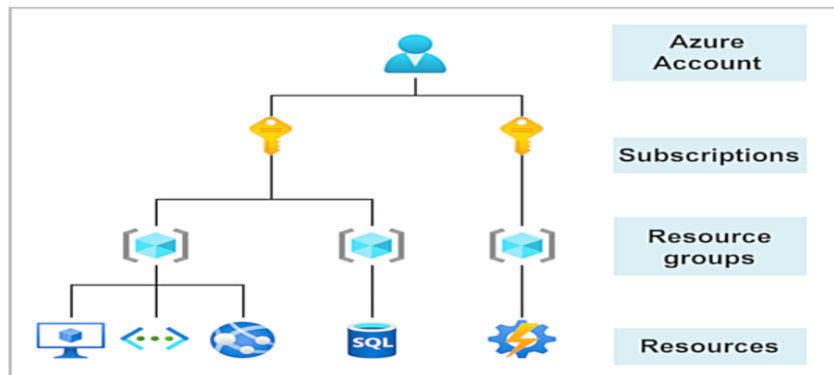
- Describe Azure regions, region pairs, and sovereign regions.
- Describe Availability Zones.
- Describe Azure datacenters.
- Describe Azure resources and Resource Groups.
- Describe subscriptions.
- Describe management groups.
- Describe the hierarchy of resource groups, subscriptions, and management groups.

Azure:

Azure is a continually expanding set of cloud services that help you meet current and future business challenges

Azure services:100+

- company might use a single Azure account for your business and separate subscriptions for development, marketing, and sales departments



Azure Free Account	Azure Student Account	Azure Account
Free access to popular Azure products for 12 months.	Free access to popular Azure products for 12 months.	purchase Azure access directly from Microsoft by signing up on the Azure website or through a Microsoft representative.
A credit to use for the first 30 days.	A credit to use for the first 12 months	
Access to more than 25 products that are always free.	Free access to certain software developer tools.	Cloud Solution Provider partners offer a range of complete managed-cloud solutions for Azure.
\$200 credits	\$100 Credits	

To interact with azure services:

1.use GUI

2.use CLI (powershell or bash)

- step 1.Choose Powershell

Powershell command: **>az version**

-step2 .To use bash command instead of choosing Bash CLI :>**bash**

- >**date**
- Try to run an update to CLI: >**az upgrade**

-step3.To switch back to powershell cmd:>**pwsh**

-step4.Unfamiliar with bash and powershell CLI but want to use it use

- > **az interactive** (with the help of this command you can autocomplete cmd)
- >**upgrade** (to update CLI)
- >**version**(to see the version)

Core architectural components of Azure :

1.physical infrastructure **2.**management infrastructure.

Interactively explore azure infra: [Global infrastructure](#)

****physical infrastructure**

***Azure Regions:**

- geographical area on the planet that contains at least one, but potentially multiple datacenters that are nearby and networked together with a low-latency network.
- global Azure services that don't require you to select a particular region, such as **Microsoft Entra ID, Azure Traffic Manager, and Azure DNS.**

***Availability zones:**

- physically separate datacenters within an Azure region
- made up of one or more datacenters equipped with independent power, cooling, and networking.
- set up to be an isolation boundary. If one zone goes down, the other continues working.

- connected through high-speed, private fiber-optic networks.
- Availability zones are primarily for **VMs, managed disks, load balancers, and SQL databases**

Important:

To ensure resiliency, a minimum of three separate availability zones are present in all availability zone-enabled regions. However, not all Azure Regions currently support availability zones.

Imp: If catastrophe (large impactful event) happens and all availability-zones in region are affected, then Azure provides another option to maintain resiliency "region-pair"

1. Zonal services: pin the resource to a specific zone (for example, **VMs, managed disks, IP addresses**).

2. Zone-redundant services: replicates automatically across zones (for example, **zone-redundant storage, SQL Database**).

3. Non-regional services: Services are always available from Azure geographies and are resilient to zone-wide outages as well as region-wide outages.

***Region-Pairs:**

- Azure regions are paired with another region within the same geography (such as US, Europe, or Asia) at least **300 miles** away.
- if a region in a pair was affected by a natural disaster, services would automatically fail over to the other region in its region pair.
- Data continues to reside within the same geography as its pair (except for Brazil South) for tax- and law-enforcement jurisdiction purposes

Important

Not all Azure services automatically replicate data or automatically fall back from a failed region to cross-replicate to another enabled region. In these scenarios, recovery and replication must be configured by the customer.

1.region-pair in two direction: eg. West US and East US back each other up

2.region-pair in one direction:eg. ---**West India's** secondary region is South India, South India does not rely on West India. West India's secondary region is South India, but South India's secondary region is Central India.---**Brazil South** is unique because it's paired with a region outside of its geography. Brazil South's secondary region is South Central US. The secondary region of South Central US isn't Brazil South.

***Sovereign Regions:**

- Sovereign regions are instances of Azure that are isolated from the main instance of Azure.

- You may need to use a sovereign region for compliance or legal purposes.
- US DoD Central, US Gov Virginia, US Gov Iowa and more: These regions are physical and logical network-isolated instances of Azure for U.S. government agencies and partners. These datacenters are operated by screened U.S. personnel and include additional compliance certifications.
- China East, China North, and more: These regions are available through a unique partnership between **Microsoft and 21Vianet**, whereby Microsoft doesn't directly maintain the datacenters.

****Management infrastructure:**

- includes **Azure resources** and **resource groups, subscriptions, and account**

***Azure resource:** basic building block of Azure.

*** Azure resource-group:** simply groupings of resources

Note:

- Some resources may be moved between resource groups, but when you move a resource to a new group, it will no longer be associated with the former group.
- If you grant or deny access to a resource group, you've granted or denied access to all the resources within the resource group.

***Azure Subscription:**

- subscriptions are a unit of management, billing, and scale
- provides you with authenticated and authorized access to Azure products and services
- An account can have multiple subscriptions, but it's only required to have one.
- In a multi-subscription account, you can use the subscriptions to configure different billing models and apply different access-management policies
- two types of subscription boundaries that you can use: 1.**billing boundary** 2. **Access control boundary**

*** Azure management groups:**

- Azure management groups provide a level of scope above subscriptions.
- All subscriptions within a management group automatically inherit the conditions applied to the management group, the same way that resource groups inherit settings from subscriptions and resources inherit from resource groups.
- Management groups give you enterprise-grade management at a large scale, no matter what type of subscriptions you might have.
- Management groups can be nested.
- 10,000 management groups can be supported in a single directory.

- A management group tree can support up to six levels of depth. This limit doesn't include the root level or the subscription level.
- Each management group and subscription can support only one parent.

Azure compute and networking services:

Learning objectives

After completing this module, you'll be able to:

- Compare compute types, including container instances, virtual machines, and functions.
- Describe virtual machine options, including virtual machines (VMs), virtual machine scale sets, virtual machine availability sets, and Azure Virtual Desktop.
- Describe resources required for virtual machines.
- Describe application hosting options, including Azure Web Apps, containers, and virtual machines.
- Describe virtual networking, including the purpose of Azure Virtual Networks, Azure virtual subnets, peering, Azure DNS, VPN Gateway, and ExpressRoute.
- Define public and private endpoints.

****Azure virtual machines:**

- Just like a physical computer, you can customize all of the software running on your VM.
- VMs are an ideal choice when you need:- 1.Total control over the operating system (OS).- 2.The ability to run custom software.- 3.To use custom hosting configurations.
- An image is a template used to create a VM and may already include an OS and other software, like development tools or web hosting environments.
- Azure can also manage the grouping of VMs for you with features such as **scale sets and availability sets**.

Virtual Machine Scale Set(VM-SS):

- Virtual machine scale sets let you create and manage a group of identical, load-balanced VMs.
- monitor the utilization to determine if you need to increase or decrease the number of VMs.
- The number of VM instances can automatically increase or decrease in response to demand, or you can set it to scale based on a defined schedule

- Virtual machine scale sets also automatically deploy a load balancer to make sure that your resources are being used efficiently
- With virtual machine scale sets, you can build large-scale services for areas such as compute, big data, and container workloads.

Virtual machine availability sets:

- Availability sets are designed to ensure that VMs stagger updates and have varied power and network connectivity, preventing you from losing all your VMs with a single network or power failure.

Availability sets accomplish these objectives by grouping VMs in two ways: **update domain and fault domain.**

Update Domain:

Explanation: In Azure, when you have many Virtual Machines (VMs) running an app, Azure does not restart all of them at the same time because that would make the app go offline.

Instead, Azure divides VMs into Update Domains (groups).

During maintenance, only one group restarts at a time, while the other groups keep running. Azure also waits (up to 30 minutes) before moving to the next group to ensure everything is healthy.

This way, the application stays available and users are not affected.

Fault Domain:

- groups your VMs by common power source and network switch.

-By default, an availability set splits your VMs across **up to three fault domains.**

- helps protect against a physical power or networking failure by having VMs in different fault domains

Note: there's no additional cost for configuring an availability set. You only pay for the VM instances you create.

Create VM with installing nginx: [Exercise - Create an Azure virtual machine - Training | Microsoft Learn](#)

****Azure Virtual Desktop:**

- Another type of virtual machine is the Azure Virtual Desktop.
- It enables you to use a cloud-hosted version of Windows from any location.
- Support windows, macOS, iOS, android
- Azure Virtual Desktop provides centralized security management for users' desktops with Microsoft Entra ID.
- With Azure Virtual Desktop, the data and apps are separated from the local hardware. The actual desktop and apps are running in the cloud, meaning the risk of confidential data being left on a personal device is reduced.

****Azure Container:**

VMs are still limited to a single operating system per virtual machine. If you want to run multiple instances of an application on a single host machine, containers are an excellent choice.

- Containers are a virtualization environment
- Much like running multiple virtual machines on a single physical host
- Unlike virtual machines, you don't manage the operating system for a container.
- Containers are lightweight and designed to be created, scaled out, and stopped dynamically
- With containers, you can quickly restart if there's a crash or hardware interruption. One of the most popular container engines is **Docker**, and Azure supports Docker.

VM :Virtualize hardware

Container:Virtualize OS.operating system level virtualization is one way .run mulitple lightweight container on single host.Container can orchestrated with container cluster orchestration.

Completely control environment:choose VM
portability,performance characterstics,mangement capablities:then choose Container

Azure Container Instances:

- simplest way to run a container in Azure without managing the vm
- (PaaS) offering.

Azure Container Apps:

- similar in many ways to a container instance.
- They allow you to get up and running right away
- remove the container management piece
- PaaS offering.
- Container Apps have extra benefits such as the ability to incorporate load balancing and scaling

Azure Kubernetes Service

- container orchestration service
- AKS can make container fleet management simpler and more efficient.

Note: Using container you can this split allows you to separate portions of your app into logical sections that can be maintained, scaled, or updated independently.

Azure functions:

- an event-driven, serverless compute option that doesn't require maintaining virtual machines or containers.
- With Azure Functions, an event wakes the function, alleviating the need to keep resources provisioned when there are no events.
- no infrastructure management
- scalability(Functions scale automatically based on demand)
- Azure only charges you for the CPU time used while your function runs.

Stateless function:When they're stateless (the default), they behave as if they restart every time they respond to an event.

Stateful function: When they're stateful (called Durable Functions), a context is passed through the function to track prior activity.

Azure App service:

- enables you to build and host web apps, background jobs, mobile back-ends, and RESTful APIs in the programming language of your choice without managing infrastructure.
- It offers automatic scaling and high availability.
- App Service supports Windows and Linux.
- app service styles like:
 - **Web apps:**App Service includes full support for hosting web apps
 - **API apps:** can build REST-based web APIs by using your choice of language and framework.Swagger support and the ability to package and publish your API in Azure Marketplace. The produced apps can be consumed from any HTTP- or HTTPS-based client.
- **WebJobs:** WebJobs feature to run a program or scripts.They can be scheduled or run by a trigger. WebJobs are often used to run background tasks as part of your application logic.

- Mobile apps: Mobile Apps feature of App Service to quickly build a back end for iOS and Android apps with just few actions:
 1. Store mobile app data in a cloud-based SQL database.
 2. Authenticate customers against common social providers, such as MSA, Google, X, and Facebook.
 3. Send push notifications.
 4. Execute custom back-end logic in C# or Node.js.

Azure Vnet:

- Azure virtual networks and virtual subnets enable Azure resources, such as VMs, web apps, and databases, to communicate with each other, with users on the internet, and with your on-premises client computers
- Provide key network capabilities:
 - Isolation and segmentation
 - Internet communications
 - Communicate between Azure resources

Point-to-site virtual private network connections are from a computer outside your organization back into your corporate network. In this case, the client computer initiates an encrypted VPN connection to connect to the Azure virtual network.

Site-to-site virtual private networks link your on-premises VPN device or gateway to the Azure VPN gateway in a virtual network. In effect, the devices in Azure can appear as being on the local network. The connection is encrypted and works over the internet.

Azure ExpressRoute provides a dedicated private connectivity to Azure that doesn't travel over the internet. ExpressRoute is useful for environments where you need greater bandwidth and even higher levels of security.

- Communicate with on-premises resources
- Route network traffic

Route tables allow you to define rules about how traffic should be directed. You can create custom route tables that control how packets are routed between subnets.

Border Gateway Protocol (BGP) works with Azure VPN gateways, Azure Route Server, or Azure ExpressRoute to propagate on-premises BGP routes to Azure virtual networks.

- Filter network traffic
Network security groups are Azure resources that can contain multiple inbound and outbound security rules. You can define these rules to allow or block traffic, based on factors such as source and destination IP address, port, and protocol.
Network virtual appliances are specialized VMs that can be compared to a hardened network appliance. A network virtual appliance carries out a particular network function, such as running a firewall or performing wide area network (WAN) optimization.
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- Connect virtual networks(vnet-peering)
 - Network traffic between peered networks is private, and travels on the Microsoft backbone network, never entering the public internet
 - Peering enables resources in each virtual network to communicate with each other.
 - virtual networks can be in separate regions.
 - This feature allows you to create a global interconnected network through Azure.

Public endpoints have a public IP address and can be accessed from anywhere in the world.

Private endpoints exist within a virtual network and have a private IP address from within the address space of that virtual network.

VM to accessible over internet we need to change NSG(allow inbound http access on port 80)

****Azure VPN:**

- A virtual private network (VPN) uses an encrypted tunnel within another network
- VPNs are typically deployed to connect two or more trusted private networks to one another over an untrusted network (typically the public internet).
- Azure VPN Gateway instances are deployed in a dedicated subnet of the virtual network and enable the following connectivity:
 - Connect on-premises datacenters to virtual networks through a site-to-site connection.
 - Connect individual devices to virtual networks through a point-to-site connection.
 - Connect virtual networks to other virtual networks through a network-to-network connection.
 - **Note:**only one VPN gateway in each virtual network.
 - setting up a VPN gateway, you must specify the type of VPN - **either policy-based or route-based**

Policy-based VPN gateways specify statically the IP address of packets that should be encrypted through each tunnel. This type of device evaluates every data packet against those sets of IP addresses to choose the tunnel where that packet is going to be sent through.

In Route-based gateways, IPsec tunnels are modeled as a network interface or virtual tunnel interface. IP routing (either static routes or dynamic routing protocols) decides which one of these tunnel interfaces to use when sending each packet. Route-based VPNs are the preferred connection method for on-premises devices. They're more resilient to topology changes such as the creation of new subnets.

- Way to maximize the resiliency of your VPN gateway:

1.Active/standby: VPN gateways are deployed as two instances in an active/standby configuration, even if you only see one VPN gateway resource in Azure.

2.Active/active: With the introduction of support for the BGP routing protocol, you can also deploy VPN gateways in an active/active configuration. you assign a unique public

IP address to each instance. You then create separate tunnels from the on-premises device to each IP address.

3. **ExpressRoute failover:** to configure a VPN gateway as a secure failover path for ExpressRoute connections.

4. **Zone-redundant gateways:** In regions that support availability zones, VPN gateways and ExpressRoute gateways can be deployed in a zone-redundant configuration. These gateways require different gateway stock keeping units (SKUs) and use Standard public IP addresses instead of Basic public IP addresses.

Azure Express route:

Describe Azure storage services

Learning objectives

Upon completion of this module, you will be able to:

- Compare Azure storage services
- Describe storage tiers
- Describe redundancy options
- Describe storage account options and storage types
- Identify options for moving files, including AzCopy, Azure Storage Explorer, and Azure File Sync
- Describe migration options, including Azure Migrate and Azure Data Box

storage account:

- provides a unique namespace for your Azure Storage data that's accessible from anywhere in the world over HTTP or HTTPS.

Redundancy option:

Redundancy in Primary Region:

1. **Locally redundant storage (LRS):**

- replicates your data three times within a single data center in the primary region.
- Provides 11 nines of durability (99.999999999%) of objects over the year
- Protects your data against server rack and drive failures

2. **Zone-redundant storage (ZRS):**

- replicates your Azure Storage data synchronously(copy at same time) across three Azure availability zones in the primary region
- Provides 12 nines of durability (99.9999999999%) of objects over a given year.
- data is still accessible for both read and write operations even if a zone becomes unavailable.
- ZRS is also recommended for restricting replication of data within a country or region to meet data governance requirements.

Redundancy in Secondary Region:

Important

Because data is replicated to the secondary region asynchronously, a failure that affects the primary region may result in data loss if the primary region can't be recovered. The interval between the most recent writes to the primary region and the last write to the secondary region is known as the recovery point objective (RPO). The RPO indicates the point in time to which data can be recovered. Azure Storage typically has an RPO of less than 15 minutes, although there's currently no SLA on how long it takes to replicate data to the secondary region.

3. **GRS:**

- GRS is similar to running LRS in two regions
- GRS copies your data synchronously three times within a single physical location in the primary region using LRS. It then copies your data asynchronously to a single physical location in the secondary region (the region pair) using LRS
- durability for Azure Storage data objects of at least 16 nines (99.99999999999999%) over a given year.

4. Geo-zone-redundant storage (GZRS):

- GZRS is similar to running ZRS in the primary region and LRS in the secondary region.
- Data in a GZRS storage account is copied across three Azure availability zones in the primary region (similar to ZRS) and is also replicated to a secondary geographic region, using LRS
- Microsoft recommends using GZRS for applications requiring maximum consistency, durability, and availability, excellent performance, and resilience for disaster recovery.
- provide at least 16 nines (99.99999999999999%) of durability of objects over a given year.
- Data in a GZRS storage account is copied across three Azure availability zones in the primary region (similar to ZRS) and is also replicated to a secondary geographic region, using LRS,

Read-access to data in secondary Region:

- Geo-redundant storage (with GRS or GZRS) replicates your data to another physical location in the secondary region to protect against regional outages
- For read access to the secondary region, enable **read-access geo-redundant storage (RA-GRS)** or **read-access geo-zone-redundant storage (RA-GZRS)**.

Important

Remember that the data in your secondary region may not be up-to-date due to RPO.

Storage Account:

Type	Supported services	Redundancy Options	Usage
Standard general-purpose v2	Blob Storage (including Data Lake Storage), Queue Storage, Table Storage, and Azure Files	LRS, GRS, RA-GRS, ZRS, GZRS, RA-GZRS	Standard storage account type for blobs, file shares, queues, and tables. Recommended for most scenarios using Azure Storage. If you want support for network file system (NFS) in Azure Files, use the premium file shares account type.

Premium block blobs	Blob Storage (including Data Lake Storage)	LRS, ZRS	Premium storage account type for block blobs and append blobs. Recommended for scenarios with high transaction rates or that use smaller objects or require consistently low storage latency.
Premium file shares	Azure Files	LRS, ZRS	Premium storage account type for file shares only. Recommended for enterprise or high-performance scale applications. Use this account type if you want a storage account that supports both Server Message Block (SMB) and NFS file shares.
Premium page blobs	Page blobs only	LRS	Premium storage account type for page blobs only.

Azure Blobs: A massively scalable object store for text and binary data. Also includes support for big data analytics through Data Lake Storage Gen2.

Azure Files: Managed file shares for cloud or on-premises deployments.

Azure Queues: A messaging store for reliable messaging between application components.

Azure Disks: Block-level storage volumes for Azure VMs.

Azure Tables: NoSQL table option for structured, non-relational data.

Storage-account tiers:

Hot access tier: Optimized for storing data that is accessed frequently (for example, images for your website).

Cool access tier: Optimized for data that is infrequently accessed and stored for at least 30 days (for example, invoices for your customers).

Cold access tier: Optimized for storing data that is infrequently accessed and stored for at least 90 days.

Archive access tier: Appropriate for data that is rarely accessed and stored for at least 180 days, with flexible latency requirements (for example, long-term backups).

Notes:

- **Hot, cool, and cold** access tiers can be set at the account level. The archive access tier isn't available at the account level.
- Hot, cool, cold, and archive tiers can be set at the blob level, during or after upload.

Azure File:

- storage offers fully managed file shares in the cloud that are accessible via the industry standard **Server Message Block (SMB) or Network File System (NFS) protocols**.
- file shares can be mounted concurrently by cloud or on-premises deployments.
- SMB Azure file shares are accessible from Windows, Linux, and macOS clients.
- NFS Azure Files shares are accessible from Linux or macOS clients.
- Additionally, SMB Azure file shares can be cached on Windows Servers with Azure File Sync for fast access near where the data is being used.

Azure Queue :storage is a service for storing large numbers of messages

Azure Disk: they're the same as a physical disk, but they're virtualized . With managed disks, all you have to do is provision the disk, and Azure will take care of the rest.

Azure Migrate is a service that helps you migrate from an on-premises environment to the cloud.

Azure Migrate: Discovery and assessment. Discover and assess on-premises servers running on VMware, Hyper-V, and physical servers in preparation for migration to Azure.

Azure Migrate: Server Migration. Migrate VMware VMs, Hyper-V VMs, physical servers, other virtualized servers, and public cloud VMs to Azure.

Data Migration Assistant. Data Migration Assistant is a stand-alone tool to assess SQL Servers. It helps pinpoint potential problems blocking migration. It identifies unsupported features, new features that can benefit you after migration, and the right path for database migration.

Azure Database Migration Service. Migrate on-premises databases to Azure VMs running SQL Server, Azure SQL Database, or SQL Managed Instances.

Azure App Service migration assistant. Azure App Service migration assistant is a standalone tool to assess on-premises websites for migration to Azure App Service. Use Migration Assistant to migrate .NET and PHP web apps to Azure.

Azure Data Box. Use Azure Data Box products to move large amounts of offline data to Azure. Azure Data Box is a physical migration service that helps transfer large amounts of data in a quick, inexpensive, and reliable way. A rugged case protects and secures the Data Box from damage during transit.

Data Box is ideally suited to transfer data sizes larger than 40 TBs in scenarios with no to limited network connectivity

Once the data from your import order is uploaded to Azure, the disks on the device are wiped clean in accordance with **NIST 800-88r1** standards. For an export order, the disks are erased once the device reaches the Azure datacenter.

Azure also has tools designed to help you move or interact with individual files or small file groups. Among those tools are **AzCopy, Azure Storage Explorer, and Azure File Sync.**

AzCopy is a command-line utility that you can use to copy blobs or files to or from your storage account.

📌 **Important**

Synchronizing blobs or files with AzCopy is one-direction synchronization. When you synchronize, you designate the source and destination, and AzCopy will copy files or blobs in that direction. It doesn't synchronize bi-directionally based on timestamps or other metadata.

Azure Storage Explorer is a standalone app that provides a graphical interface to manage files and blobs in your Azure Storage Account

Azure File Sync is a tool that lets you centralize your file shares in Azure Files and keep the flexibility, performance, and compatibility of a Windows file server. Configure cloud tiering so the most frequently accessed files are replicated locally, while infrequently accessed files are kept in the cloud until requested

Introduction to Cloud Infrastructure: Describe Azure management and governance

Learning objectives

After completing this module, you'll be able to:

- Describe factors that can affect costs in Azure.
- Compare the Pricing calculator and Total Cost of Ownership (TCO) calculator.
- Describe the Microsoft Cost Management Tool.
- Describe the purpose of tags.

TCO: helps to understand the saving of operating system solution on azure compare to the on-premises data center

Pricing Calculator: helps you to determine the which azure services is best fit for your budget

Azure Advisor: To monitor your actual cost. making recommendation on unused resources. way to optimize the service and also set spending limits to prevent accidental cost overruns

Capital expense (CapEx) : building out and maintaining infrastructure and facilities

Operational expense (OpEx) : renting infrastructure as you need it, whether it's compute, storage, networking, and so on.

That OpEx cost can be impacted by many factors. Some of the impacting factors are:

- Resource type
- Consumption
- Maintenance
- Geography
- Network Traffic:

Billing zones are a factor in determining the cost of some Azure services.

Bandwidth refers to data moving in and out of Azure datacenter

he [bandwidth pricing page](#) has additional information on pricing for data ingress, egress, and transfer.

- Subscription type
- Azure Marketplace

Note: **The Total Cost of Ownership (TCO) calculator has been retired**

Cost Management provides the ability to quickly check Azure resource costs, create alerts based on resource spend, and create budgets that can be used to automate management of resources.

Cost analysis is a subset of Cost Management that provides a quick visual for your Azure costs.

Cost alert : single location where all alerts we can see
three types of alerts that may show up are:

1. Budget alerts
2. Credit alerts
3. Department spending quota alerts.

Budget alerts notify you when spending, based on usage or cost, reaches or exceeds the amount defined in the alert condition of the budget. Cost Management budgets are created using the **Azure portal** or the **Azure Consumption API**.

- An alert email is also sent to the people in the alert recipients list of the budget.

Credit alerts notify you when your Azure credit monetary commitments are consumed.. Monetary commitments are for organizations with Enterprise Agreements (EAs).

- credit alerts are generated automatically at **90% and at 100%** of your Azure credit balance.
- Whenever an alert is generated, it's reflected in cost alerts, and in the email sent to the account owners.

Department spending quota alerts notify you when department spending reaches a fixed threshold of the quota. Spending quotas are configured in the EA(enterprise agreement portal)

- Whenever a threshold is met, it generates an email to department owners, and appears in cost alerts. For example, 50 percent or 75 percent of the quota.

A budget is where you set a spending limit for Azure.

Resource tags are another way to organize resources. Tags provide extra information, or metadata, about your resources.

- **Resource management**
- **Cost management and optimization**
- **Operations management**
- **Security**
- **Governance and regulatory compliance**
- **Workload optimization and automation**

- You can use **Azure Policy** to enforce tagging rules and conventions.
- Resources don't inherit tags from subscriptions and resource groups
- A resource tag consists of a name and a value

Additional link: [Control Azure spending and manage bills with Microsoft Cost Management + Billing](#)

Describe features and tools in Azure for governance and compliance

Learning objectives

After completing this module, you'll be able to:

- Describe the purpose of Microsoft Purview.
- Describe the purpose of Azure Policy.
- Describe the purpose of resource locks.
- Describe the purpose of the Service Trust portal.

***Microsoft Purview :

- family of data governance, risk, and compliance solutions that helps you get a single, unified view into your data
- brings insights about your on-premises, multicloud, and software-as-a-service data together.
- With Microsoft Purview, you can stay up-to-date on your data landscape thanks to:
 1. Automated data discovery
 2. Sensitive data classification
 3. End-to-end data lineage (**Data Lineage**: It provides a detailed roadmap of how data flows through systems, enabling transparency, trust, and better decision-making.)

- Two main solution areas comprise Microsoft Purview: **risk and compliance** and **unified data governance**.
- **Microsoft 365 features** as a core component of the Microsoft Purview risk and compliance solutions.
- Microsoft Purview's robust data governance capabilities enable you to manage your data stored in Azure, SQL and Hive databases, locally, and even in other clouds like Amazon S3.

***Azure Policy:

- is a service in Azure that enables you to create, assign, and manage policies that control or audit your resources.
- enables you to define both **individual policies** and **groups of related policies**, known as initiatives.
- evaluates your resources and highlights resources that aren't compliant with the policies you've created.
- can also prevent noncompliant resources from being created.
- set policies on a specific resource, resource group, subscription and inherited
- comes with built-in policy and initiative definitions for Storage, Networking, Compute, Security Center, and Monitoring
- If you have a specific resource that you don't want Azure Policy to automatically fix, you can flag that resource as an exception – and the policy won't automatically fix that resource.
- Azure Policy also integrates with Azure DevOps by applying any continuous integration and delivery pipeline policies that pertain to the pre-deployment and post-deployment phases of your applications.

An **Azure Policy initiative** is a way of grouping related policies together .e.g. Enable Monitoring in Azure Security Center

Resource lock

- prevents resources from being accidentally deleted or changed.
- Resource locks can be applied to individual resources, resource groups, or even an entire subscription.
- Resource locks are inherited

- Type:
- **Delete** means authorized users can still read and modify a resource, but they can't delete the resource.
- **ReadOnly** means authorized users can read a resource, but they can't delete or update the resource. Applying this lock is like restricting all authorized users to the permissions granted by the Reader role.

***Service Trust portal:

- Microsoft Service Trust Portal is a portal that provides access to various content, tools, and other resources about Microsoft security, privacy, and compliance practices.
- You can access the Service Trust Portal at <https://servicetrust.microsoft.com/>.
- Service Trust Portal reports and documents are available to download for at least 12 months after publishing or until a new version of document becomes available.

Additional link: [Intro to Azure Policy](#)

Learning objectives

After completing this module, you'll be able to:

- Describe the Azure portal.
- Describe Azure Cloud Shell, including Azure CLI and Azure PowerShell.
- Describe the purpose of Azure Arc.
- Describe Azure Resource Manager (ARM), ARM templates, and Bicep.

Azure portal is a web-based, unified console that provides an alternative to command-line tools

Azure Cloud Shell is a browser-based shell tool that allows you to create, configure, and manage Azure resources using a shell.

Azure Cloud Shell support both Azure PowerShell and the Azure Command Line Interface (CLI), which is a Bash shell.

Azure PowerShell is a shell with which developers, DevOps, and IT professionals can run commands called command-lets (cmdlets)

- can install and configure Azure PowerShell on Windows, Linux, and Mac platforms.

Azure CLI is functionally equivalent to Azure PowerShell, with the primary difference being the syntax of commands

- use bash command
- installable on Windows, Linux, and Mac platforms

Azure Arc lets you extend your Azure compliance and monitoring to your hybrid and multi-cloud configurations. Azure Arc simplifies governance and management by delivering a consistent multi-cloud and on-premises management platform.

Azure Arc allows you to manage the following resource types hosted outside of Azure:

- Servers
- Kubernetes clusters
- Azure data services
- SQL Server
- Virtual machines (preview)

Azure Resource Manager (ARM) :

is the deployment and management service for Azure. It provides a management layer that enables you to create, update, and delete resources in your Azure account. Anytime you do anything with your Azure resources, ARM is involved.

- Manage your infrastructure through declarative templates rather than scripts.

Infrastructure as code is a concept where you manage your infrastructure as lines of code.

- **ARM templates** and **Bicep** are two examples of using infrastructure as code with the Azure Resource Manager to maintain your environment.

ARM template then orchestrates the creation of those resources in parallel. That is, if you need 50 instances of the same resource, all 50 instances are created at the same time.

Bicep is a language that uses declarative syntax to deploy Azure resources.

- **Support for all resource types and API versions:** Bicep immediately supports all preview and GA versions for Azure services

Additional link: [Implement resource management security](#)

Learning objectives

By the end of this module, you'll be able to:

- Describe the purpose of Azure Advisor.
- Describe Azure Service Health.
- Describe Azure Monitor, including Azure Log Analytics, Azure Monitor Alerts, and Application Insights.

Azure Advisor:

- evaluates your Azure resources and makes recommendations to help improve reliability, security, and performance, achieve operational excellence, and reduce costs

Reliability is used to ensure and improve the continuity of your business-critical applications.

Security is used to detect threats and vulnerabilities that might lead to security breaches.

Performance is used to improve the speed of your applications.

Operational Excellence is used to help you achieve process and workflow efficiency, resource manageability, and deployment best practices.

Cost is used to optimize and reduce your overall Azure spending.

Azure Service Health: helps you keep track of Azure resource, both your specifically deployed resources and the overall status of Azure. Azure service health does this by combining three different Azure services:

- Azure Status: broad picture of the status of Azure globally
- Service Health :broad picture of the status of Azure globally
- Resource Health: tailored view of your actual Azure resource

Azure Monitor is a platform for collecting data on your resources, analyzing that data, visualizing the information, and even acting on the results.

Azure Monitor Alerts are an automated way to stay informed when Azure Monitor detects a threshold being crossed

Application Insights, an Azure Monitor feature, monitors your web applications. two ways to configure Application Insights :either install an SDK in your application, or you can use the Application Insights agent. The Application Insights agent is supported in C#.NET, VB.NET, Java, JavaScript, Node.js, and Python.

- can also configure it to periodically send synthetic requests to your application, allowing you to check the status and monitor your application even during periods of low activity.

Additional link:

[Get started with Azure Advisor](#)

[Intro to Azure Service Health i](#)

[Monitor the usage, performance, and availability of resources with Azure Monitor](#)

